

ECE 1387

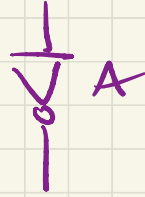
Lecture 11

Nov 28, 2025

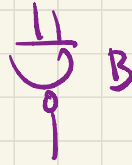
- Technology mapping, LP/ILP
- Quiz next week
- Course evaluations
- A3 return next week

Gate

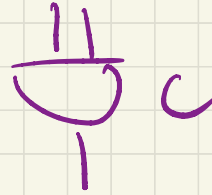
NOT



NAND



AND



OR



area

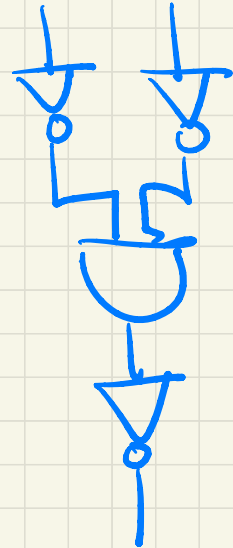
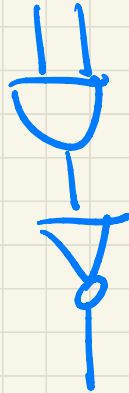
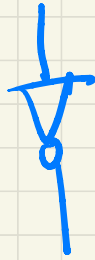
0.5

1

1.2

1.2

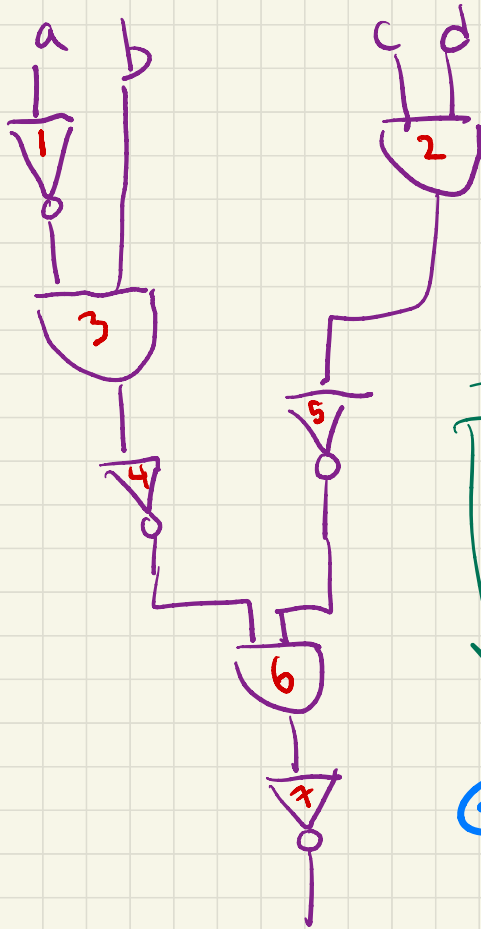
Pattern graph



BN:

$$f = \bar{a}b + cd$$

BN



@1 A matches
cost = 0.5

@2 C matches
cost = 1.2

@3 C matches
cost = 1.2 + cost@1
0.5
= 1.7

@4 A matches, cost
0.5 + cost@3
= 0.5 + 1.7 = 2.2

$$\begin{aligned} \text{B matches cost} &= 1 + \underbrace{\text{cost @ 1}}_{0.5} \\ &= 1.5 \end{aligned} \quad \text{Best}$$

$$\begin{aligned} \text{@ 5 A matches cost} &= 0.5 + \underbrace{\text{cost @ 2}}_{1.2} \\ &= 1.7 \end{aligned}$$

$$\text{B matches cost} = 1$$

@ 6 C matches cost =

$$1.2 + \underset{1.5}{\text{cost@4}} + \underset{1}{\text{cost@5}} \\ = 3.7$$

@ 7 A matches cost =

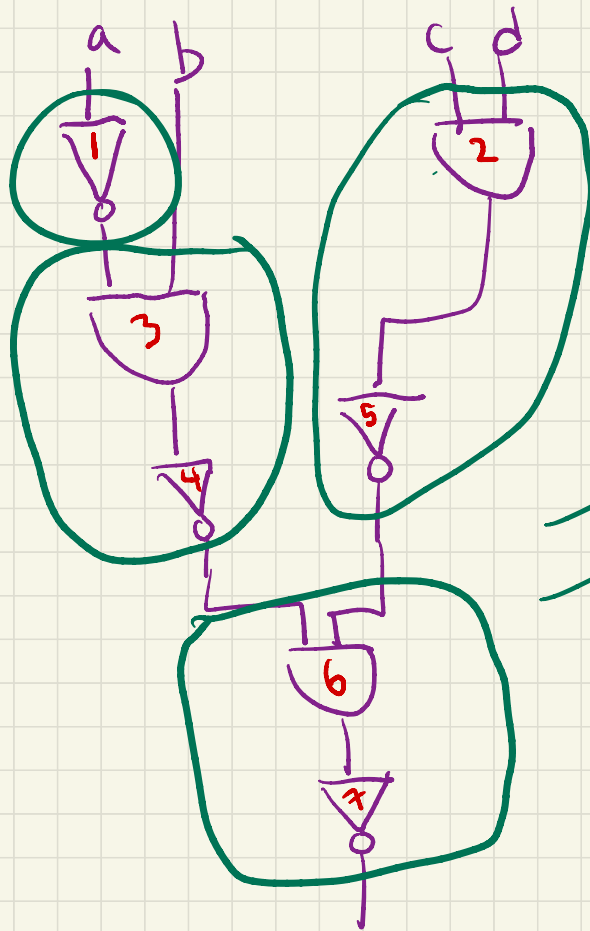
$$0.5 + \underbrace{\text{cost@6}}_{3.7} \\ = 4.2$$

B matches cost =

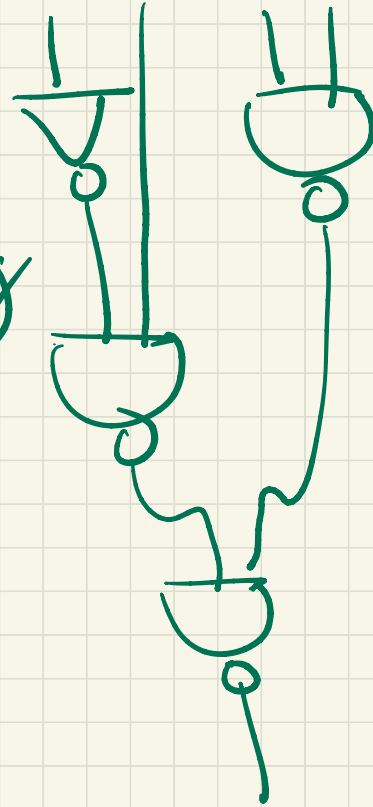
$$1 + \underset{1.5}{\text{cost@4}} + \underset{1}{\text{cost@5}} \\ = 3.5$$

D matches cost =

$$1.2 + \underset{1.7}{\text{cost@3}} + \underset{1.2}{\text{cost@2}} \\ = 4.1$$



mapping



Summary of Algorithm

1) Traverse subject graph from inputs to output (topological order)

For each node visited, find all matches of pattern graphs @ node
For each match, determine cost (based on gate costs +

inputs to gate cost)

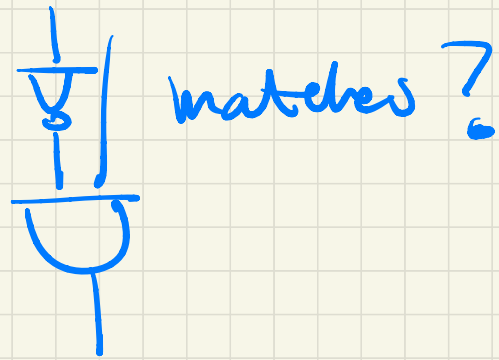
- Choose best match (with lowest cumulative cost), store in table
-

2) Traverse subject graph in reverse order (outputs to inputs), construct mapping

Also works for delay, just
carry forward slowest path

Sub-problem: matching

— how to check if



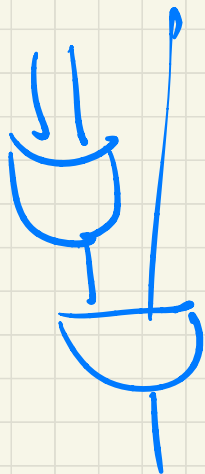
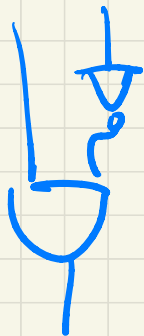
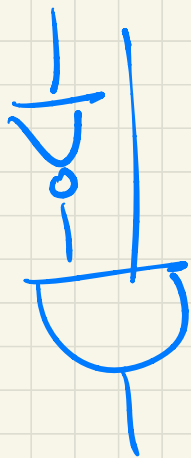
— tree matching

"Structural"

→ walk up both trees (from a node or subgraph)

→ and compare

→ have a match if finish with no mismatch.



for such
asymmetric
patterns, can
make matching
aware of
swappable inputs,
OR can put multiple
variants into
pattern graph lib.

I also Boolean matching

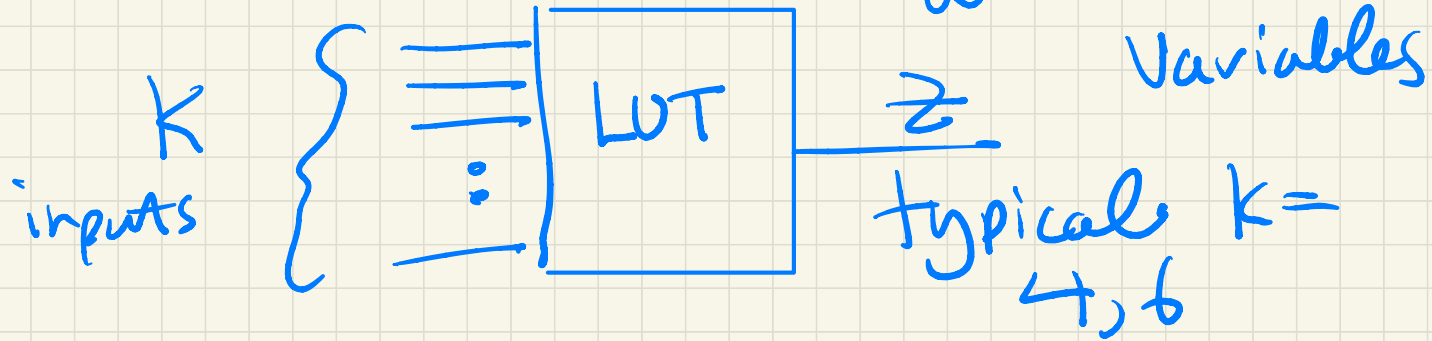
↳ compare logic functions
of std cell gates and
subgraphs of subject graph

FPGAs

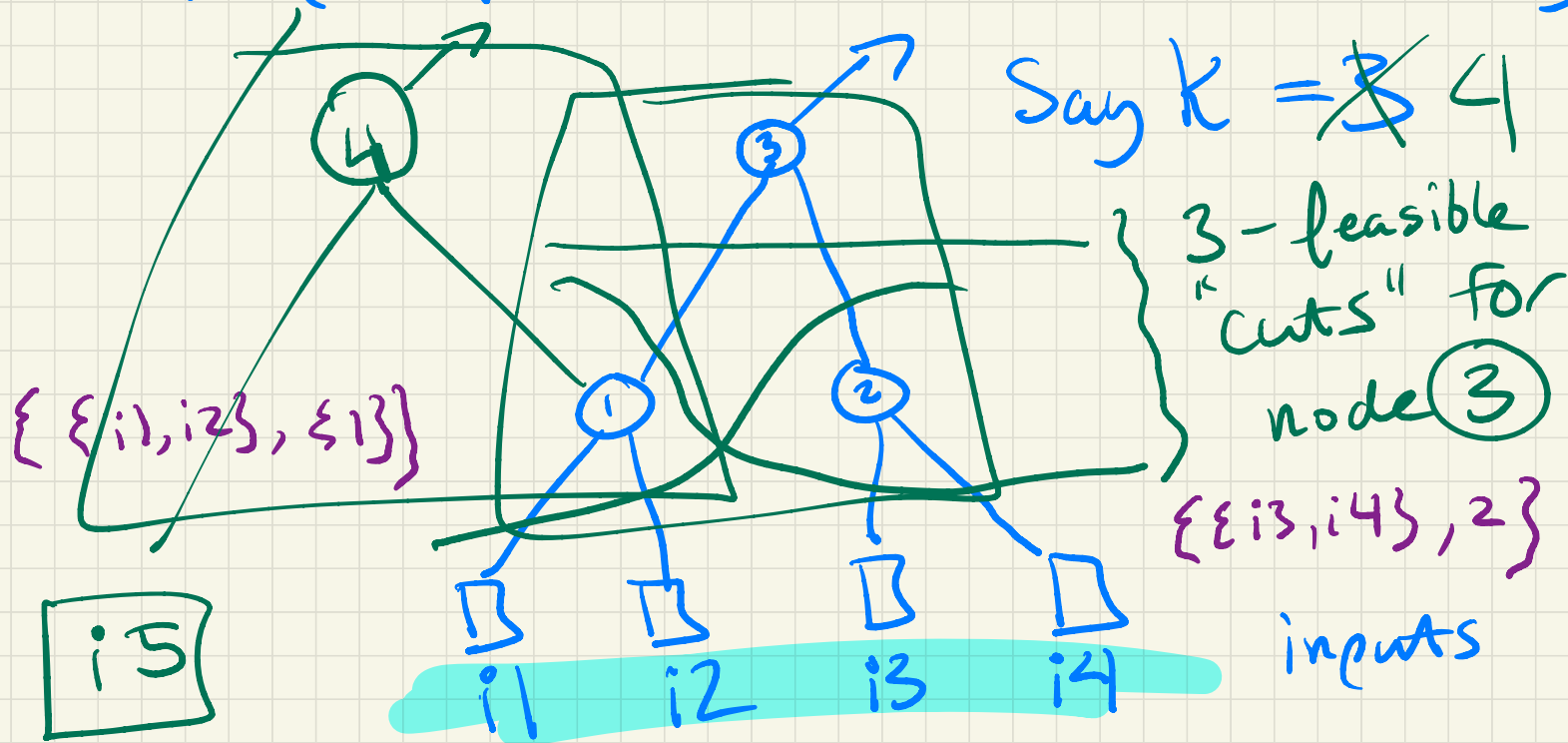
- Same basic dynamic programming tech mapping is used

- Matching is a bit different

- K-LUT



Matching only needs to care about
of inputs (not logic functions)



Each 3-feasible cut is a "match"
→ can be implemented in a 3-LUT
a "cut" can be characterized
by its inputs

3-feasible cuts for (3)

$\left\{ \begin{array}{l} \{1, 2\}, \{i1, i2, 2\}, \\ \{1, i3, i4\} \end{array} \right\}$

How to find cuts for v

Let A, B be two sets of cuts

$\nearrow$
a cut: set of nodes
producing signals
crossing cut

$$A \diamond B = \{ p \cup q \mid p \in A, q \in B, |p \cup q| \leq k \}$$

Set cuts for v , $\phi(v)$
with fans $n_1, n_2, \dots, n_m$

$$\phi(v) = \left\{ \phi(n_1) \diamond \phi(n_2) \dots \diamond \phi(n_m) \right. \\ \left. + \{v\} \right\}$$

$\nearrow$
trivial cut.

Can find K -feasible cuts for
node v by combining cuts
from v 's fanins.

(Integer) linear programming

ILP / LP

$$f(x_1, x_2, \dots, x_n)$$

↙ want to
minimize or
maximize

$$= c_1 \cdot x_1 + c_2 \cdot x_2 + \dots + c_n \cdot x_n$$

} obj.
function

Subject to constraints

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

⋮

if all x_i 's are Real, LP } Poly time
if all x_i 's are Int, ILP } NP-hard

Joe writes	15 lines / hour
Sally "	20 lines / hour
Joe tests	25 lines / hour
Sally "	15 lines / hours

Joe, Sally work 8 hours a day

How to allocate Joe's
Sally's time?

JT: # hours Joe tests

JW: " " Joe writes

ST:

SW:

Objective function

$$\max 15JW + 20SW$$

Subject to:

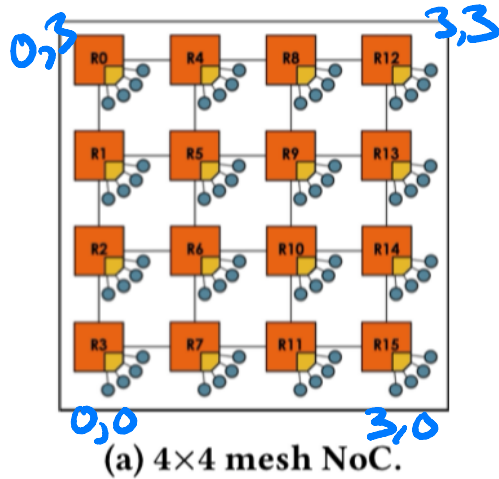
$$JT + JW = 8$$

$$ST + SW = 8$$

$$15JW + 20SW = 25JT + 15ST$$

$$JT \geq 0, JW \geq 0, ST \geq 0, SW \geq 0$$

NoC (Network on Chip) Placement



HEART'24

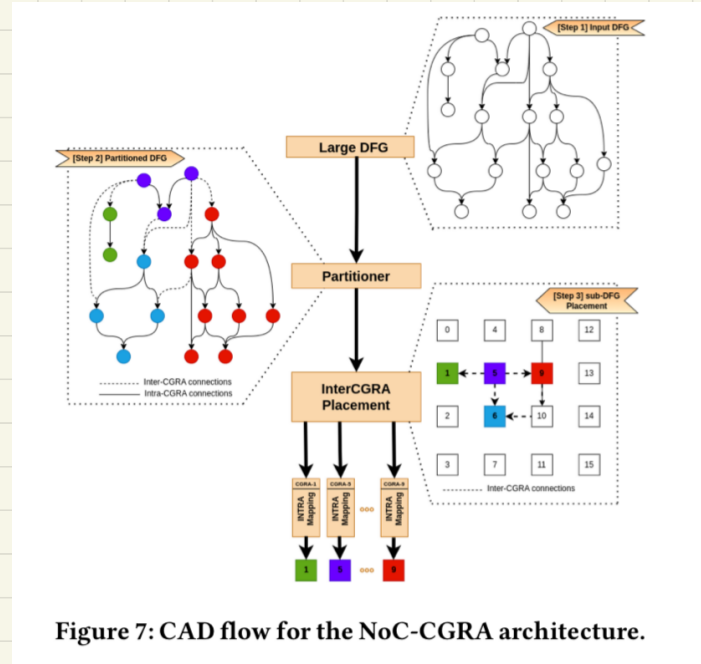


Figure 7: CAD flow for the NoC-CGRA architecture.

$\forall i \in \text{Partition}$ x_i, y_i position of
partition i in NoC mesh

$\forall i, j \in \text{Partitions} : r_{x_{i,j}}, r_{x_{j,i}},$
 $r_{y_{i,j}}, r_{y_{j,i}}$
binary variables

$\forall n \in \text{Connections} : l_{x_n}, h_{x_n}, l_{y_n}, h_{y_n}$
are integer variables representing
lowest x , highest x , low y , high y for
conn n .

min

$$\phi = \sum_{n \in \text{Conns}} (h x_n - l x_n) + (h y_n - l y_n)$$

↳ HPWL

Subject to:

$x_i \geq 0$ $x_i \leq 3$
 $y_i \geq 0$ $y_i \leq 3$ } stay inside
NoC mesh

$$\forall i, j \in \text{Partitions}, i \neq j : x_i < x_j + r_{x_{ij}} \cdot M$$

$$,, : x_j < x_i + r_{x_{ji}} \cdot M$$

$$,, : y_i < y_j + r_{y_{ij}} \cdot M$$

$$,, : y_j < y_i + r_{y_{ji}} \cdot M$$

$$\forall i, j \in \text{Partitions}, i \neq j$$

$$r_{x_{ij}} + r_{x_{ji}} + r_{y_{ij}} + r_{y_{ji}} \leq \underline{3}$$